WORKED SOLUTION

STEP II 1988, Question 14

Mechanics • Rotating Rigid Bodies

STEP II 1988, Question 14

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/88-s2-q14

The key idea

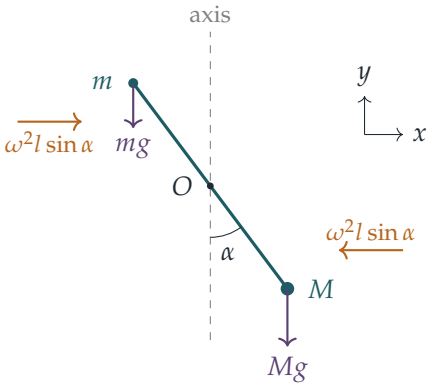
In the steady state nothing moves relative to the rotating vertical plane that contains the rod, so each particle simply goes round a horizontal circle at constant speed. That fixes each particle's acceleration completely. Newton's second law then tells us the force the rod exerts on each particle, and two facts about a *light* rod finish the job: the forces on it must balance, and (because it swings freely about the axle) their moments about the axle must balance too.

Solution

Model. Treat the masses as particles on a light rigid rod, pivoted smoothly at O so that the axle exerts no couple about its own direction. Work in the vertical plane containing the rod, which turns with angular speed ω about the vertical line through O. Take x horizontal in this plane and y vertically up, with origin O.

Suppose M is at $(l\sin\alpha,\ -l\cos\alpha)$ and m is at $(-l\sin\alpha,\ l\cos\alpha)$; we shall see in a moment that the heavier particle must be the lower one. Each particle moves on a horizontal circle of radius $l\sin\alpha$ about the vertical axis at constant speed, so its acceleration is $\omega^2 l\sin\alpha$ horizontally towards the axis:

$$\mathbf{a}_M = (-\omega^2 l\sin\alpha,\ 0), \qquad \mathbf{a}_m = (\omega^2 l\sin\alpha,\ 0).$$



(Amber arrows are accelerations, plum arrows are weights.)

Forces from the rod. Let $\mathbf{R}_M$ and $\mathbf{R}_m$ be the forces the rod exerts on M and m. These need not lie along the rod: a rigid rod can also push sideways. Newton's second law gives $\mathbf{R}_M - Mg\,\mathbf{j} = M\mathbf{a}_M$, so

$$\mathbf{R}_M = \big(-M\omega^2 l\sin\alpha,\ Mg\big), \qquad \mathbf{R}_m = \big(m\omega^2 l\sin\alpha,\ mg\big).$$

By Newton's third law the particles push on the rod with $-\mathbf{R}_M$ at M's end and $-\mathbf{R}_m$ at m's end.

Ask yourself The axle exerts an unknown force on the rod at O. How can we take moments without knowing it?

Moments about the axle. The axle's force acts at O, so it has no moment about O; and the rod swings freely, so the axle supplies no couple about its own direction (the line through O perpendicular to our plane). A light rod has no angular momentum of its own, so the moments of the two end forces about this line must cancel. The moment of a force (F_x, F_y) acting at (x, y) is $xF_y - yF_x$ (anticlockwise). For $-\mathbf{R}_M$ at $(l\sin\alpha, -l\cos\alpha)$:

$$l\sin\alpha\,(-Mg) - (-l\cos\alpha)(M\omega^2 l\sin\alpha) = -Mgl\sin\alpha + M\omega^2 l^2\sin\alpha\cos\alpha,$$

and for $-\mathbf{R}_m$ at $(-l\sin\alpha,\ l\cos\alpha)$:

$$(-l\sin\alpha)(-mg) - l\cos\alpha\,(-m\omega^2 l\sin\alpha) = mgl\sin\alpha + m\omega^2 l^2\sin\alpha\cos\alpha.$$

Setting the sum to zero and dividing by $l\sin\alpha \neq 0$ (as $\alpha > 0$):

$$(M+m)\,\omega^2 l\cos\alpha = (M-m)\,g \quad\Longrightarrow\quad \omega^2 = \left(\frac{M-m}{M+m}\right)\frac{g}{l\cos\alpha},$$

dividing by $\cos\alpha > 0$, as required.

If instead M were the *upper* particle, the same calculation (swap M and m) would give $(M+m)\omega^2 l\cos\alpha = (m-M)g < 0$, which is impossible. So the heavier particle is indeed below O, as assumed.

The reaction on the axle. The rod is light, so the total force on it is zero: the axle pushes on it with $\mathbf{R}_M + \mathbf{R}_m$, and hence the rod pushes on the axle with

$$\mathbf{F} = -(\mathbf{R}_M + \mathbf{R}_m) = \Big((M-m)\,\omega^2 l \sin\alpha,\ -(M+m)\,g \Big).$$

This points downwards, with horizontal part directed outwards on M's side. If ϕ is its angle with the downward vertical, then, using the value of ω^2,

$$\begin{aligned}\tan\phi &= \frac{(M-m)\,\omega^2 l \sin\alpha}{(M+m)\,g} = \frac{M-m}{M+m} \cdot \frac{l\sin\alpha}{g} \cdot \frac{M-m}{M+m} \cdot \frac{g}{l\cos\alpha} \\ &= \left(\frac{M-m}{M+m}\right)^2 \tan\alpha.\end{aligned}$$

Since $0 \le \phi < \frac{1}{2}\pi$, $\phi = \tan^{-1}\Big[\big(\frac{M-m}{M+m}\big)^2 \tan\alpha\Big]$, which is the required result.

Checks

- If $m = 0$ the system is a conical pendulum of length l, and the formula gives the familiar $\omega^2 = g/(l\cos\alpha)$. The reaction angle is then α: the rod pulls along itself, like a string.
- If $M = m$ the rod is balanced at O, so $\omega = 0$ and the reaction is vertical, with size $2Mg$; both formulae agree.
- Numerically, with $M = 3$, $m = 1$, $l = 0.7$, $\alpha = 0.6$ and $g = 9.8$ we get $\omega^2 \approx 8.48$ and a reaction angle of about 9.71°. Stepping the equation of motion for the rod's angle from rest at α keeps it at α, and computing the particles' accelerations from their positions in three dimensions reproduces the same angle.

Common slips

- Assuming the rod's forces on the particles act along the rod. They don't: a rigid rod transmits sideways forces, and that is why we take moments.
- Forgetting that m's acceleration is towards the axis too, so on the *opposite* side from M's. The horizontal parts of the reaction partly cancel, giving $M - m$ rather than $M + m$.
- Putting the heavier particle on top. The torque equation shows this cannot happen.

Beyond STEP

The rod's angular momentum about O is not parallel to the vertical spin axis, so it turns as the system rotates. Changing it needs a couple, which the axle supplies in a direction perpendicular to the one we used. This is the start of rigid-body dynamics with the

inertia tensor, and it explains why badly balanced wheels wobble on their bearings. Kleppner and Kolenkow, *An Introduction to Mechanics*, treats this carefully.

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